

WALEED BIN KHALID (Senior Software Engineer)

Lahore, Pakistan | [linkedin.com/in/waleed-bin-khalid-296b931b4](https://www.linkedin.com/in/waleed-bin-khalid-296b931b4)

+92 309 4184587 | waleedbinkhalid48@gmail.com

SUMMARY

Software Engineer with 5+ years of experience delivering full-stack web and SaaS platforms, now operating in a senior engineering capacity — owning architecture decisions, technical leadership, and end-to-end delivery. Includes 4 years building production systems with Python, Django, and Django REST Framework paired with React/Next.js on the frontend, plus additional experience delivering full-stack products on the MERN stack (MongoDB, Express, React, Node.js) and Laravel. Comfortable owning both sides of the stack: backend architecture, relational database design (PostgreSQL), query optimization, caching, and authentication/authorization, alongside modern, performant React/Next.js frontends. Experienced integrating AI features including RAG (Retrieval-Augmented Generation) pipelines, speech-to-text and text-to-speech (STT/TTS), and NLP, as well as third-party AI model APIs for tasks such as document parsing and workflow automation. Experienced leading development teams and client communication across projects for clients in the UAE, Netherlands, Canada, and Saudi Arabia. Passionate about solving real-world problems through clean architecture and efficient code.

WORK EXPERIENCE

Argon

Senior Software Engineer

Arfa Tower, Lahore, Pakistan

Dec 2023 – Present

- **Projects:**
 - FitNee (fitnee.fit) — UAE-based online fitness platform.
 - Easytipping (easytipping.com) — tipping and invoicing platform.
 - Abodems (myabode.soft2develop.com) — stay/booking management website for vacation rentals.
 - Housekeeper (admin.housekeeper.host) — property management website.
 - Mark Lauren (djf.marklauren.com) — fitness application with customizable workouts and exercises.
 - eFileMy1099 (efilemy1099.com) — IRIS tax filing website for 1099 forms.
- Built and maintained RESTful APIs using Python, Django, and Django REST Framework (DRF) for core backend services, running alongside Node.js/Express services across the Argon project portfolio.
- Designed Django ORM models and PostgreSQL schemas, and optimized queries and indexing to improve data retrieval performance for high-traffic platforms.
- Used Celery for background task processing and asynchronous integrations, including AI model calls and document-processing workflows.
- Built a RAG (Retrieval-Augmented Generation) application, exposed through a Django/DRF API layer, to power an internal employee polling and feedback system.
- Developed STT and TTS functionality for a learning-game backend using Amazon Transcribe and ElevenLabs models.
- Integrated AI models via third-party APIs for multiple tasks, including automated document parsing and workflow automation.
- Extended Django's built-in authentication and permissions system and built Django admin tooling for internal operations and content management.
- Developed and managed a UAE-based online fitness platform with 20+ trainers, 50 trainees, and 25+ nutritionists in the first 2 months.
- Proven experience implementing Stripe, PayPal, and Saudi-based HyperPay payment gateways for payment processing using Apple Pay, Google Pay, ACH, and credit cards.
- Front-End stack: React.js, Next.js, TypeScript, HTML, CSS, SASS, Tailwind, Reactstrap, Redux Toolkit, RTK Query for API implementation, Interceptor.
- Back-End stack: Python, Django, Django REST Framework, Celery, Node.js, Express.js, PostgreSQL.
- Built and secured backend services with authentication/authorization, input validation, and rate limiting to protect against common API vulnerabilities.
- Experience with AWS cloud services, including EC2 for deployment, S3 for storage, RDS for databases, and Amplify.
- Led frontend and backend teams, scheduling daily scrum calls to ensure a seamless development process.
- Utilized code splitting, lazy loading, and profiling techniques to avoid extra re-rendering and enhance frontend performance.
- Implemented advanced data-fetching strategies using RTK Query to optimize API calls and improve data handling.

- Designed and maintained scalable and secure RESTful APIs (Django REST Framework and Express) to support high-traffic applications.
- Employed server-side rendering (SSR) and static site generation (SSG) with Next.js to improve SEO and initial load times.
- Conducted performance tuning and optimization for Django and Node.js backend services to handle high volumes of traffic and data.
- Wrote unit tests with Pytest (Django) and Jest (frontend), plus end-to-end tests with Playwright, to catch regressions and ensure application reliability.

BoreDM

boredm.com — Remote

Software Engineer

Feb 2023 – Nov 2023

- Contributed to a subsurface data management platform, building the frontend in React and TypeScript and backend data services in Python/Django.
- Built Django-based data-processing services and REST API endpoints to power React-PDF-based reporting.
- Designed Django models and PostgreSQL schema for subsurface dataset storage and retrieval.
- Focused on data visualization and React-PDF-based reporting.
- Maintained direct communication with clients for requirement gathering and delivery.

The Webiestorms Company

Johar Town, Lahore, Pakistan

Software Engineer

2020 – Jan 2023

- Developed web platforms for Netherlands-based businesses, including YRH and Centropix.
- Worked closely with a Canadian firm for 1 year as a React developer.
- Actively worked as a key stakeholder to identify bottlenecks or future complications before system design.
- Followed SDLC processes, best code management practices, and state-of-the-art methodologies and design patterns.
- Implemented caching mechanisms to improve data retrieval speeds and reduce server load.
- Optimized database queries and indexing strategies to enhance application performance.
- Followed MVC structure and RESTful API development for web applications.
- Utilized project management tools such as JIRA, Monday, and Trello.
- Front-End stack: React.js, HTML, CSS, SASS, jQuery, Redux Toolkit.
- Back-End stack: Node.js, Laravel MVC Framework, SQL.
- Managed multiple projects, handled client communication, and led a team of 5 developers.

SKILLS

Backend: Python, Django, Django REST Framework (DRF), Celery, Node.js, Express.js, PostgreSQL, SQL, Laravel, Database Design & Modeling, Query Optimization & Indexing, Caching Strategies, Authentication & Authorization

Frontend: React.js, Next.js, TypeScript, HTML, CSS, SASS, Tailwind, Reactstrap, Redux Toolkit, RTK Query

APIs & Tools: RESTful APIs (Swagger, Postman), JIRA, Trello, Monday

Testing: Pytest (Django unit testing), Jest (frontend unit testing), Playwright (end-to-end testing)

Cloud/DevOps: AWS (EC2, S3, RDS, Amplify)

AI/ML Integration: RAG, NLP, STT/TTS (Amazon Transcribe, ElevenLabs), OpenAI-style model APIs, automated document parsing

Soft Skills: Problem Solving, Analytical Reasoning, Team Leadership, Client Communication, Verbal/Non-Verbal Communication

EDUCATION

Punjab University College of Information Technology, Lahore

Bachelor of Software Engineering

Johar Town, Lahore, Pakistan

July 2021